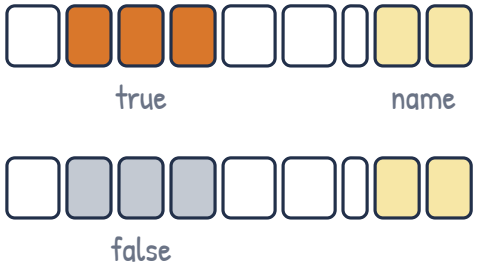
Reading the method figures

1 2 3 steps, in order

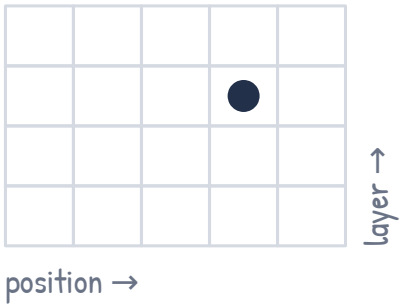
Prompt tokens



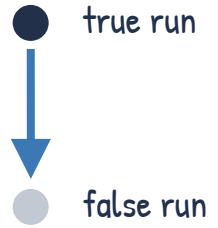
Layers



State = dot



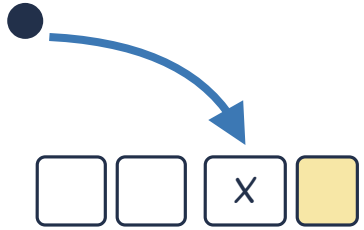
Copy a state



Block



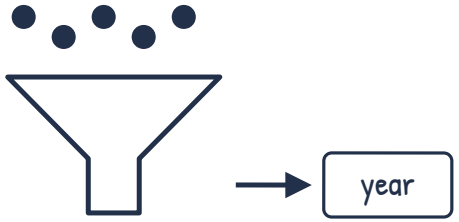
Inject



Read out



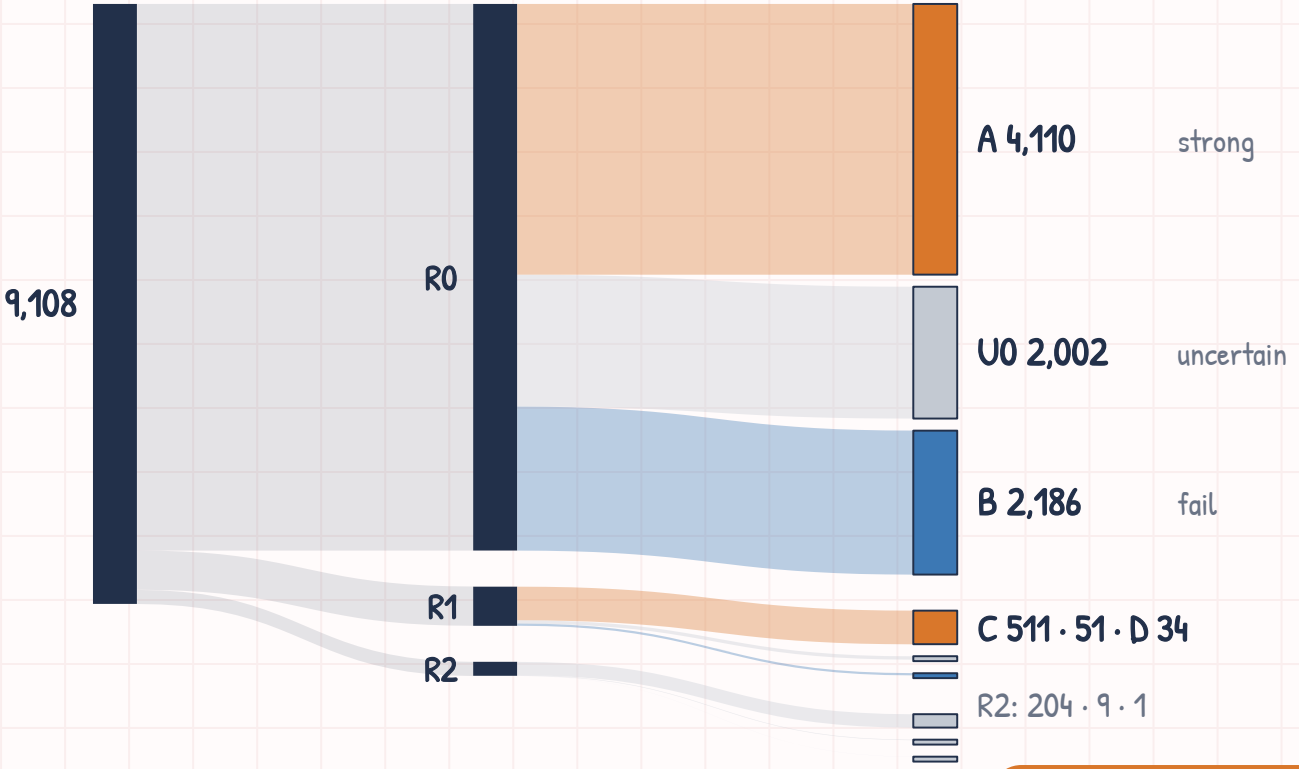
Probe



Groups and samples

example: cell C

1 papers → 2 open recall → 3 recognition → 4 cells → 5 samples



Stage 10 · M2 · M3

all papers

Stages 11–13 · X

593 target + 593 control

1,178 matched pairs

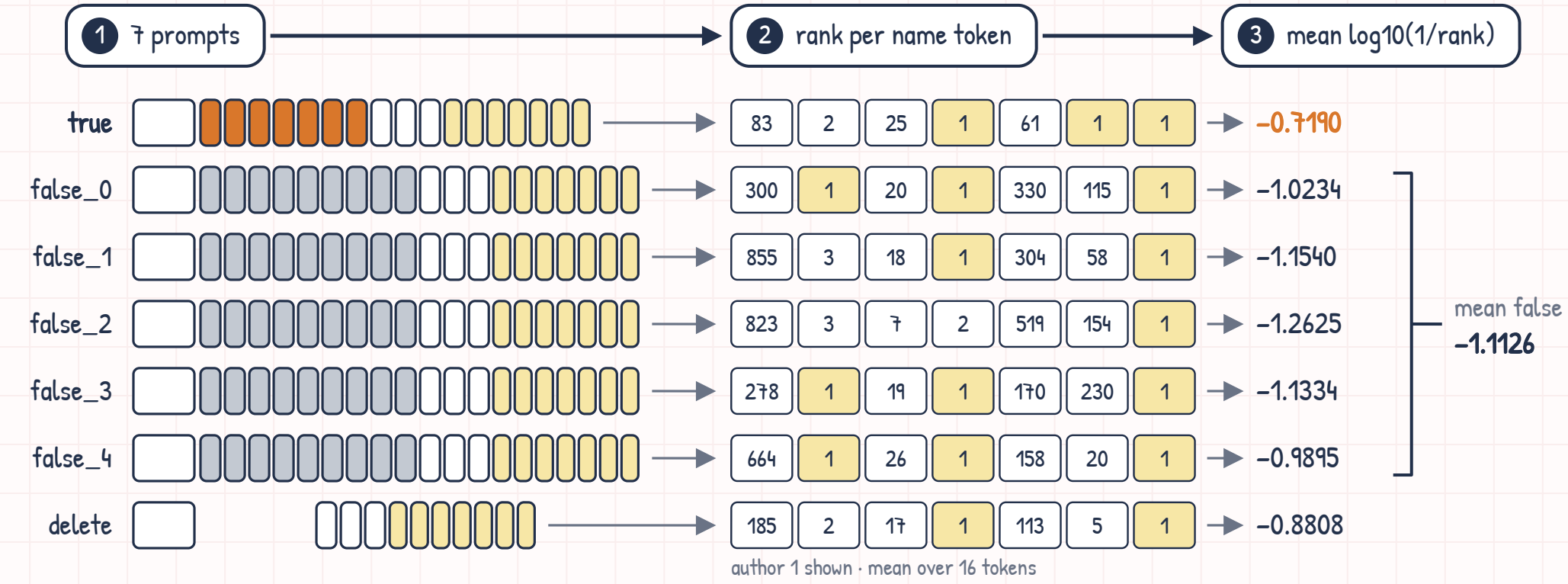
Stages 1–9

Pilot50 · Stage 8 (A, B)

target = A + C = 4,621 · control = B + D = 2,220

Scoring: rank → log → spec

Coifman, Bonanno, Rafaeli (2007) · output layer



4 contrast

spec = mean nlr(true) − mean over 5 false titles

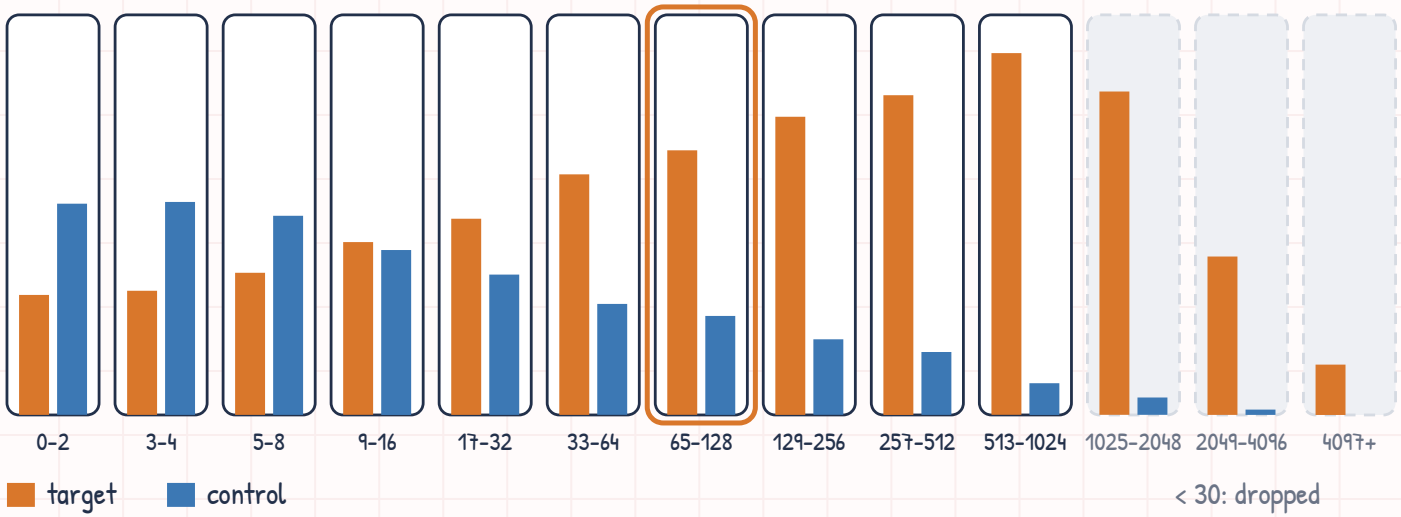
= −0.7190 − (−1.1126) = +0.3935

total = true − delete = +0.1618 · topic = mean false − delete = −0.2318

Statistical conventions

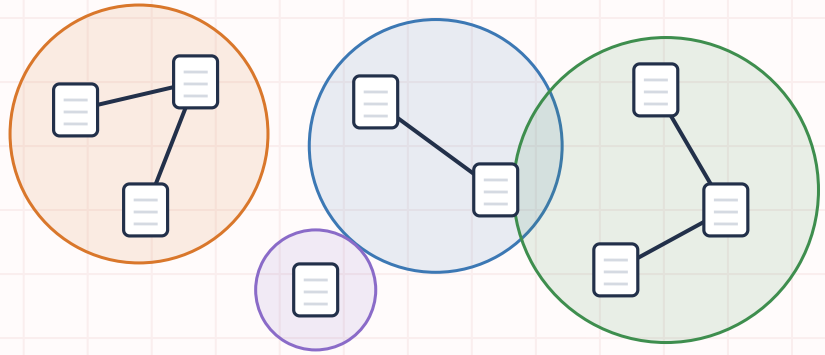
example paper: bin 65–128

1 citation bins → 2 target − control → 3 weighted mean



$$\Delta = \Sigma\, w_b \cdot (\text{mean}_t - \text{mean}_c), \quad w_b \propto n_t \cdot n_c / (n_t + n_c)$$

4 teams → 5 resample teams



shared author name = one team

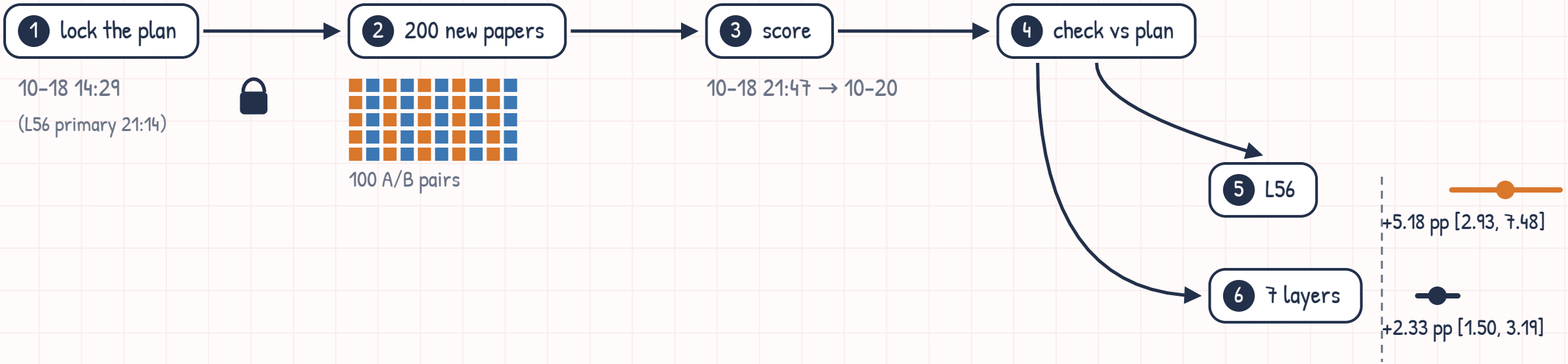
7,247 teams



one team, one fold · 2,000 draws

Independent replication (Stage 8)

plan first, data second



RQ1 = win rate(true title) − mean win rate(20 false titles)

first author vs 100 false names · J-Lens · paper bootstrap

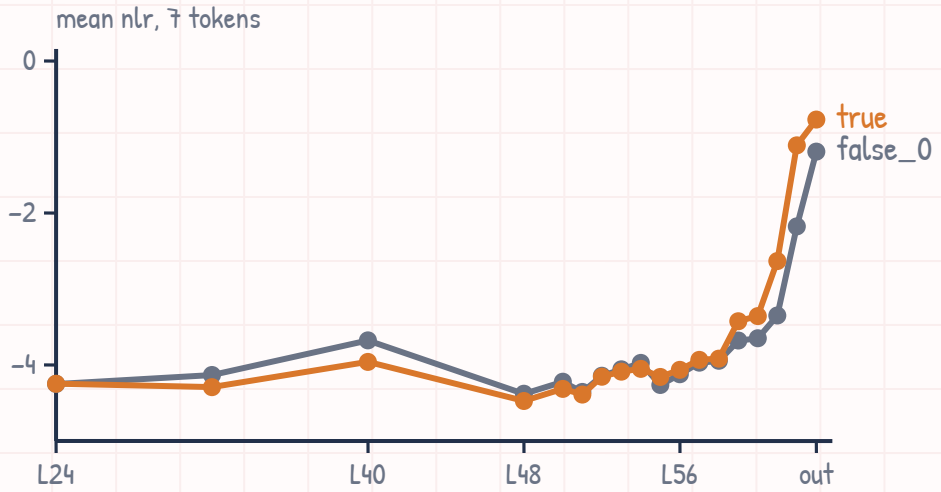
Layer-by-layer readout

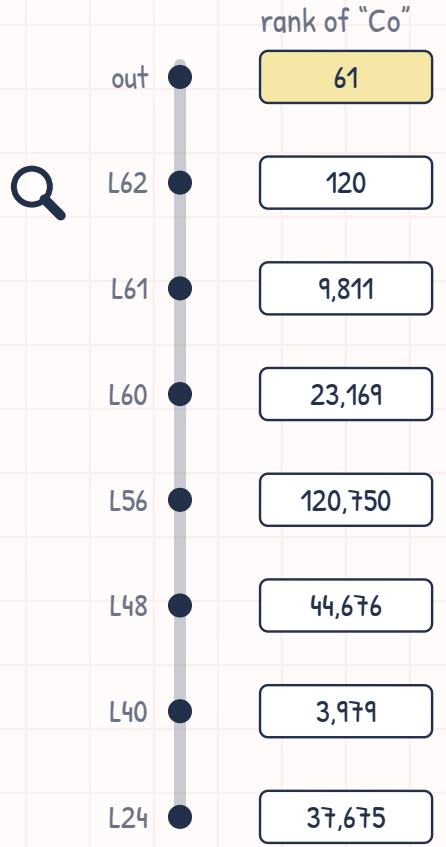
Coifman, Bonanno, Rafaeli (2007) · logit lens

1 prompt → 2 state at name → 3 lens → vocabulary → 4 rank of true token → 5 mean

nlr = log10(1 / rank)

rank = 1 + #{tokens scored above the true one}



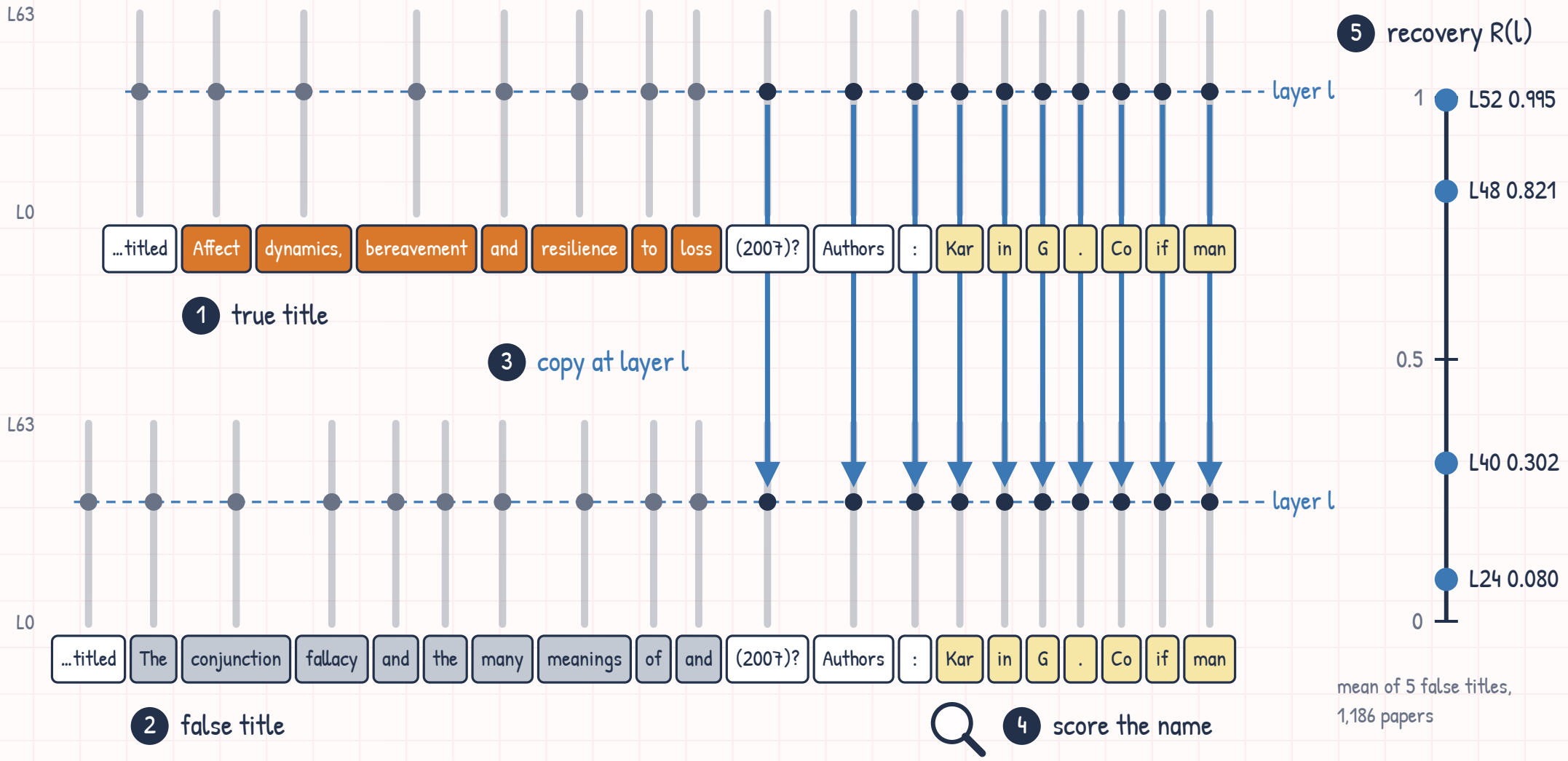


…titled | Affect | dynamics, | … | (2007)? | Authors | : | Kar | in | G | . | Co | if | man

Replacement (patching)

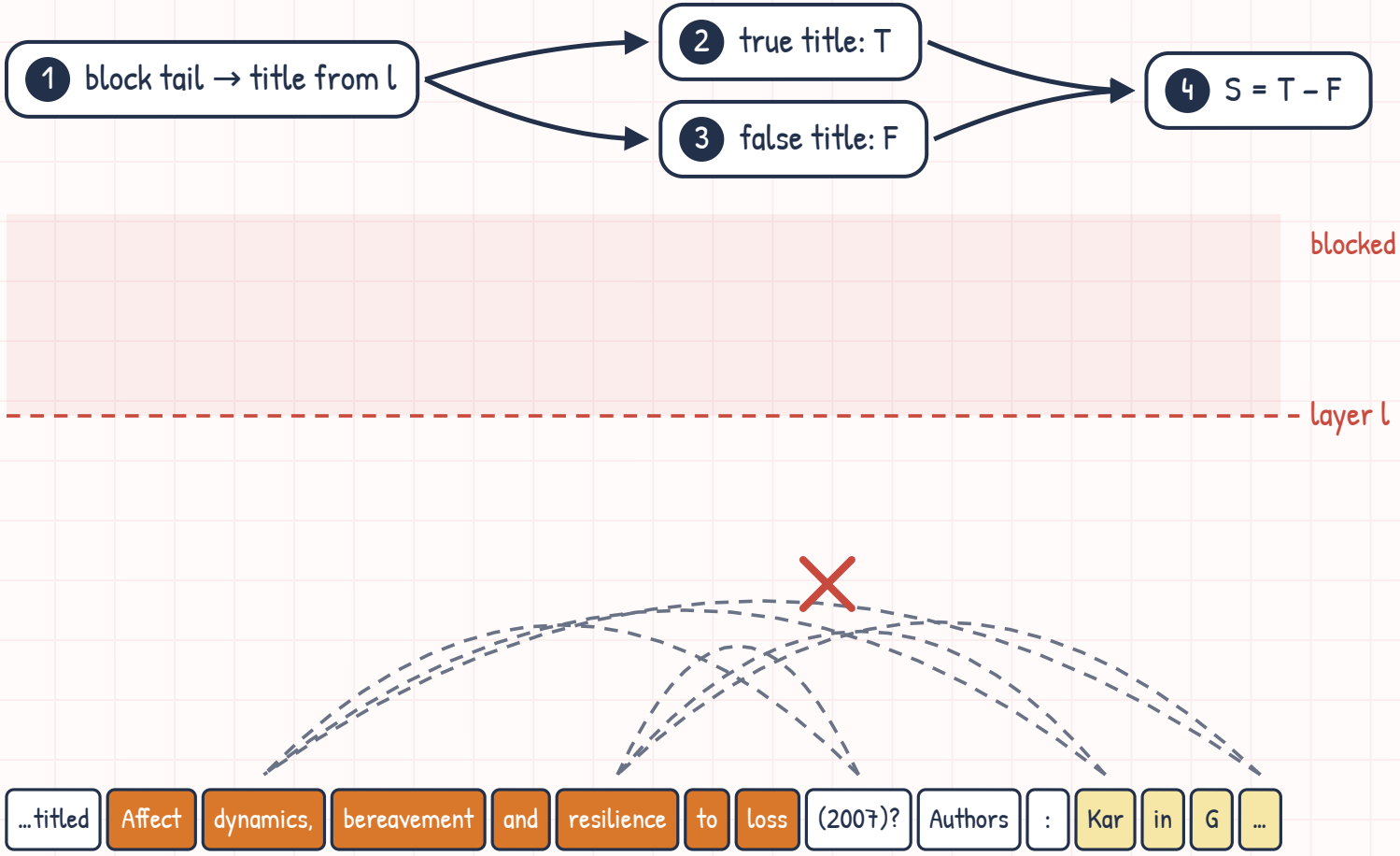
Coifman, Bonanno, Rafaeli (2007)

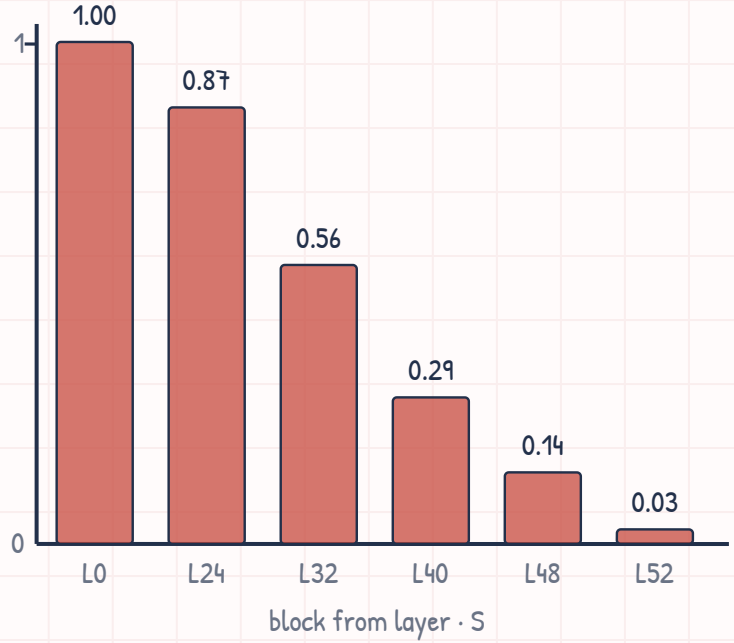
$$R(l) = (\text{patched} - \text{false}) \div (\text{true} - \text{false})$$



Knockout (block attention)

Coifman, Bonanno, Rafaeli (2007)





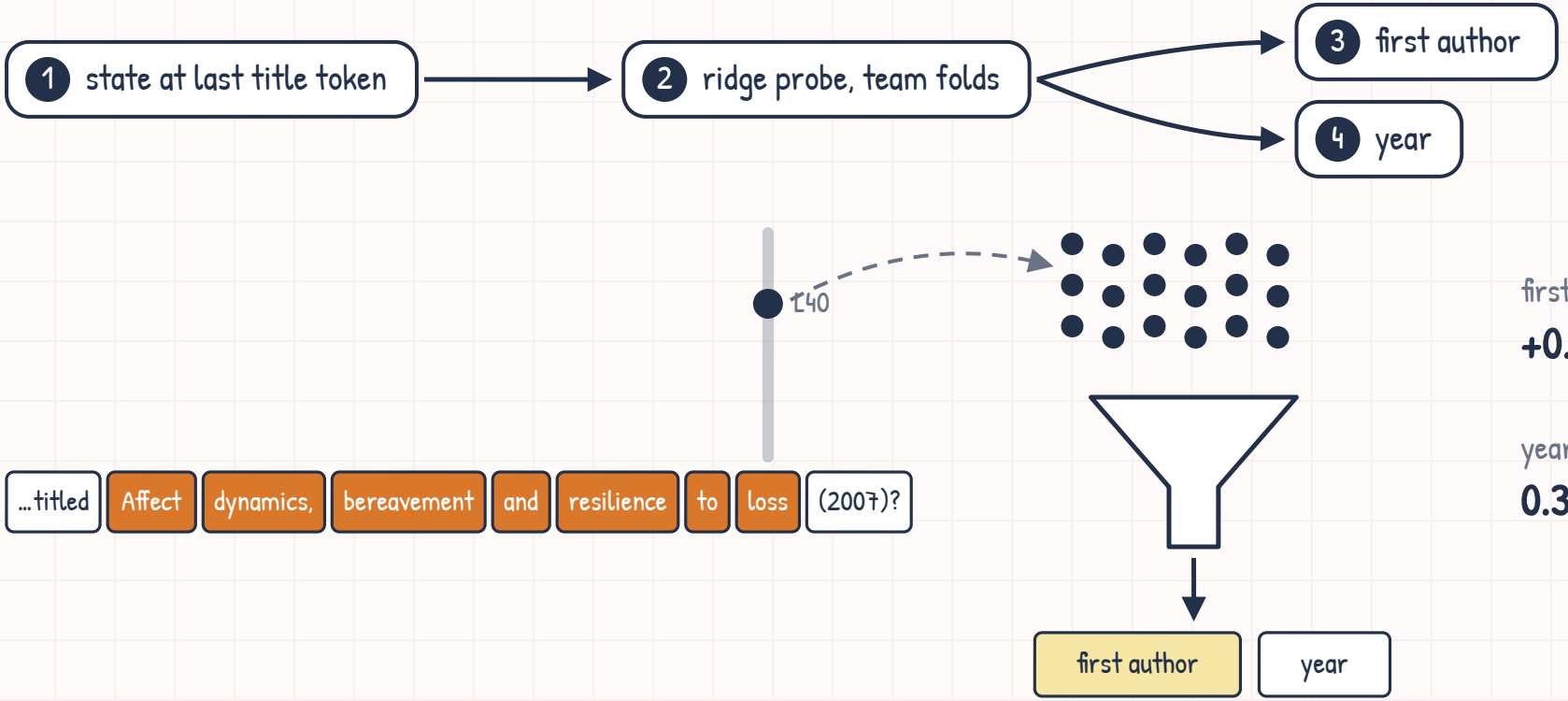
L52: target 0.018 · control 0.051

S(l) = T(l) − F(l)

loss as share of (true − false), 1,186 papers

Probe

Coifman, Bonanno, Rafaeli (2007) · state at "loss"



first author, L40

+0.061

year R^2, L16 → L52

0.38 → 0.56



colour = author team; one team, one fold

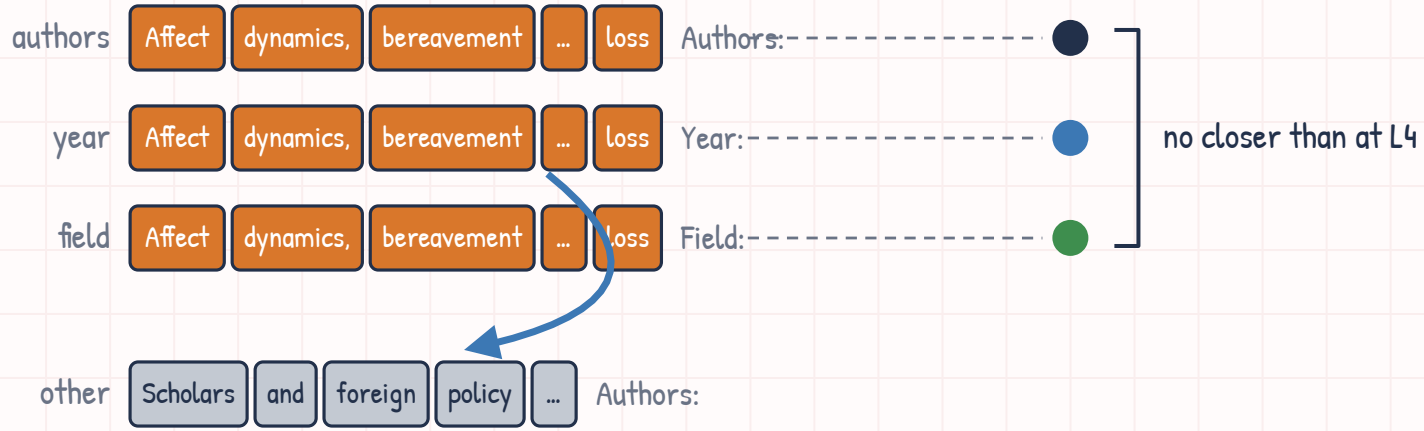
spec = probe(true title) − probe(false title)

probe score = log10(1/rank) of the true first-author token

Changing the question

Coifman, Bonanno, Rafaeli (2007)

1. three questions
2. M3: compare states
3. X1: copy into other title
4. score first author



R from	authors	year	field
L24	0.99	0.98	0.99
L40	0.92	0.92	0.92
L48	0.42	0.41	0.42

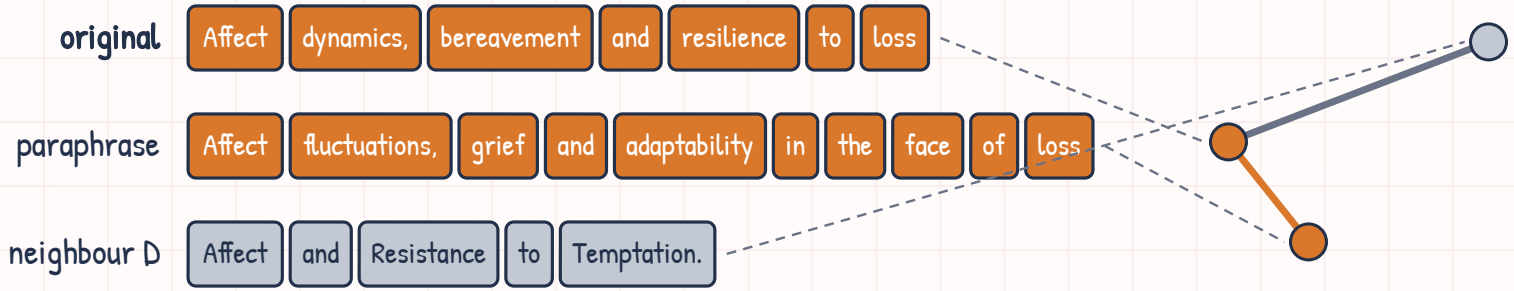
$$R(l) = (\text{patched} - \text{other}) \div (\text{own} - \text{other})$$

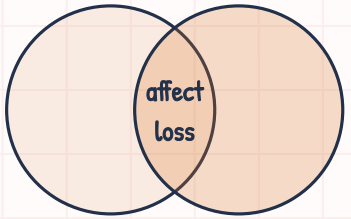
title states copied at layer l; 582 test papers

Paraphrased title (X3)

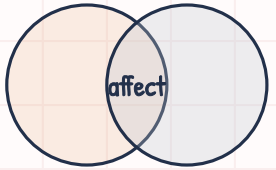
same paper in other words vs other paper, same words

1 three titles → 2 states at last title token → 3 compare distances → 4 word overlap





para 0.22



D 0.14

G′ = cos(orig, para) − cos(orig, D)

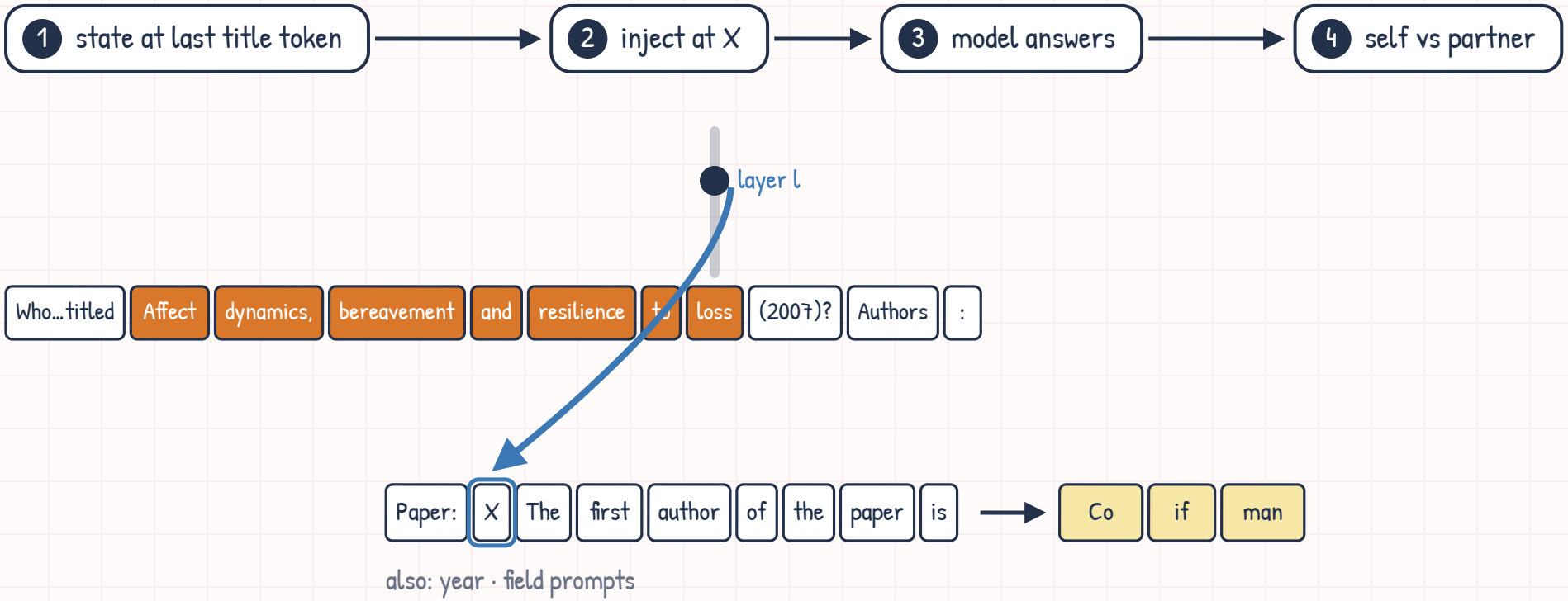
+0.32 at L8 · peak +0.46 at L36 (test set)

medians: 0.61 vs 0.15

first-author score, paraphrase − D: +0.239

Patchscope (X4)

Coifman, Bonanno, Rafaeli (2007)



D = score(own state) − score(partner state)

score = log10(1/rank) of the true answer tokens

first author from L8: +0.046

year from L40: +0.013

Group comparisons

example paper: cell C

1. 9 cells
2. P1: A – B
3. P2: C – A

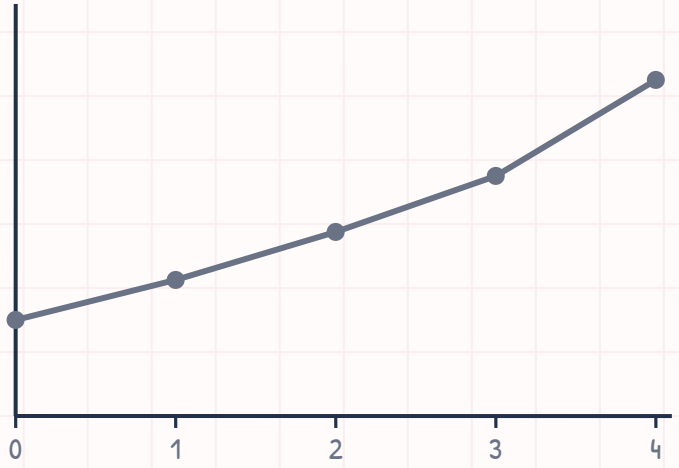
4. dose: orders right

P1

P2	strong	uncertain	fail
R0	A 4,110	U0 2,002	B 2,186
R1	C 511	51	D 34
R2	204	9	1

P1 Δ_1

P2 Δ_2



schematic, not results · 示意，不是结果

$$\text{contrast} = \Sigma\, w_b\, (\text{mean_cell1} - \text{mean_cell2})$$

title-specific signal, unwritten authors, output layer; team bootstrap